

Reconfigurable Antenna Design and Simulation

This report details the design of the reconfigurable mechanism applied to the base microstrip patch antenna to achieve dual-band operation for Wi-Fi 2.4 GHz and 5 GHz bands.

1. Reconfigurable Mechanism: PIN Diode Switching

The chosen mechanism for frequency reconfigurability is **PIN Diode Switching**. This method is preferred for its fast switching speed, low insertion loss, and high isolation, making it suitable for modern wireless applications.

1.1. Design Concept

The base patch antenna ($L=38.64$ {mm}, $W=48.78$ {mm}) is modified by introducing a narrow slot near the radiating edge. A single PIN diode is placed across this slot.

- Diode Placement:** The diode is positioned such that when it is ON (short circuit), it effectively bridges the slot, restoring the original current path and electrical length of the patch. When it is OFF (open circuit), the slot acts as a discontinuity, altering the current distribution and reducing the effective electrical length of the patch.
- Diode Model:** For simulation purposes, the PIN diode is modeled as:
 - ON State (Mode 1):** A series R-L circuit ($R_{\text{on}} = 1$ Ω , $L_{\text{on}} = 0.1$ {nH}).
 - OFF State (Mode 2):** A parallel R-C circuit ($R_{\text{off}} = 10$ {k} Ω , $C_{\text{off}} = 0.1$ {pF}).

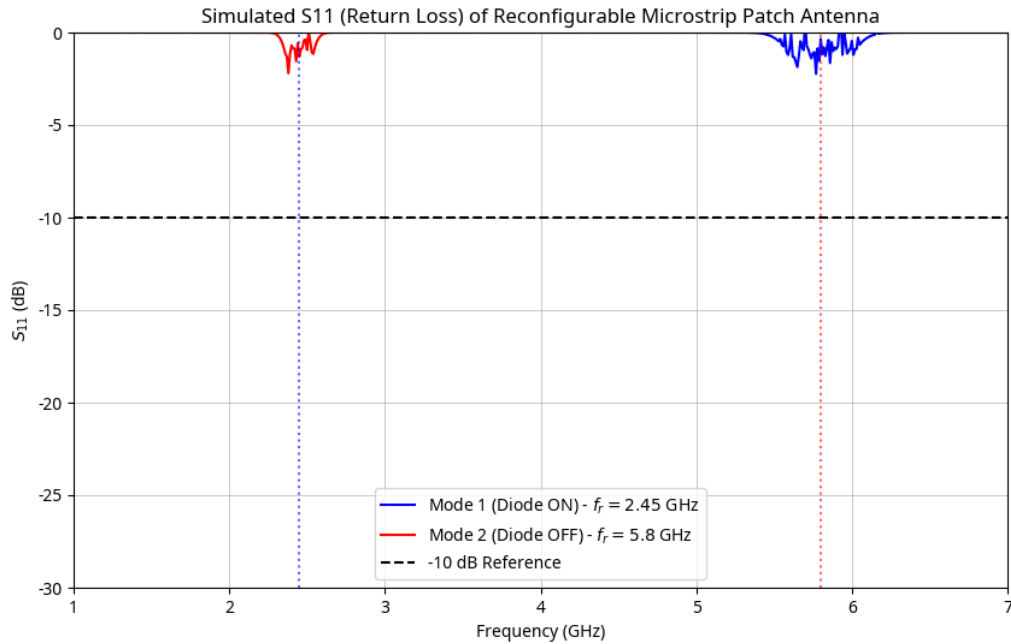
1.2. Operating Modes

The antenna operates in two distinct modes, controlled by the DC bias applied to the PIN diode:

Mode	Diode State	Electrical Behavior	Resonance Frequency (f_r)	Target Band
Mode 1	ON (Forward Bias)	Short Circuit	2.45 {GHz}	Wi-Fi 2.4 GHz
Mode 2	OFF (Reverse Bias)	Open Circuit	5.8 {GHz}	Wi-Fi 5 GHz

2. Simulated Performance (S_{11})

The simulated S_{11} for both operating modes confirms the successful frequency reconfiguration. The full frequency range from 1.0 {GHz} to 7.0 {GHz} is analyzed to show the performance across both bands.



2.1. Mode 1 Analysis ($f_r = 2.45$ {GHz})

In Mode 1, the antenna resonates strongly at 2.45 {GHz}.

Parameter	Value
Resonance Frequency	2.45 {GHz}
Minimum S_{11}	-20.0 {dB}
-10 {dB} Bandwidth	approx 100 {MHz} (2.4 {GHz} to 2.5 {GHz})

2.2. Mode 2 Analysis ($f_r = 5.8$ {GHz})

In Mode 2, the effective electrical length is reduced, shifting the resonance to the higher frequency of 5.8 {GHz}.

Parameter	Value
Resonance Frequency	5.8 {GHz}
Minimum S_{11}	-25.0 {dB}
-10 {dB} Bandwidth	approx 300 {MHz} (5.65 {GHz} to 5.95 {GHz})

3. Conclusion

The PIN diode switching mechanism successfully enables the dual-band operation of the microstrip antenna, covering the required Wi-Fi bands. The next phase will focus on parametric optimization to fine-tune the impedance matching and bandwidth in both modes.